

Chapter 33 — Urologic Aspects of Pediatric Nephrology — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **CAKUT** (congenital anomalies of kidney and urinary tract) = commonest cause of paediatric ESKD (end-stage kidney disease).
- **Neonatal creatinine** = maternal for 48 h; own by day 7 (<0.5 mg/dL term).
- **Microscopic haematuria**: >5 red cells/high-power field; no routine voiding cystourethrogram (VCUG) or cystoscopy.
- **Nephrotic**: urine protein:creatinine ratio (UPCR) >2.0 mg/mg (or >40 mg/m²/h) + low albumin + hyperlipidaemia.
- **No biopsy before steroids**; biopsy steroid-resistant (SRNS; no remission 6–8 weeks) → focal segmental glomerulosclerosis (FSGS).
- **Low C3, normal C4** → post-infectious GN (PIGN)/C3 glomerulopathy; **low C3 + C4** → lupus/membranoproliferative GN (MPGN).
- **Atypical haemolytic uraemic syndrome (aHUS)** → eculizumab; Shiga toxin *E. coli* (STEC+) HUS → supportive, avoid platelets.
- **Type 4 renal tubular acidosis (RTA) from obstructive uropathy** → urological surgery.
- **Acute kidney injury (AKI)**: serum creatinine (SCr) ↑ ≥0.3 mg/dL/48 h or ≥1.5× or urine output (UO) <0.5 mL/kg/h ≥6 h.
- **Transplant only after** ≥10 kg, 65 cm and lower-tract reconstruction.

RENAL FUNCTION AND ELECTROLYTE HOMEOSTASIS

- **Renal blood flow**: 3–4% → 12–16% cardiac output in year 1.
- **GFR (glomerular filtration rate)**: doubles by 2 weeks; term ≈40; adult by 2 years.
- **Total body water** 70–90% in neonates.
- **Neonatal free-water loss** 1–2% body weight/day, first 5 days.
- **Dilute to 50 mOsm/L**; concentrating ability limited.
- **Na⁺**: negative week 1 (atrial natriuretic peptide) → positive by 2–3 weeks.
- **K⁺ higher** in neonates: aldosterone high, tubule unresponsive.
- **Bicarbonate low**; matures over 3–4 weeks; proximal tubule reabsorbs 80%.

HAEMATURIA

- **Isolated microscopic** ≈1% school-age; <25% persist; most resolve by 1 year.
- **Definitive diagnosis** <30% microscopic; majority of gross.
- **Commonest cause of gross haematuria**: urinary tract infection.
- **Tea-coloured + RBC casts** → glomerular; red → non-glomerular.
- **Heme-positive, no RBCs** → haemoglobinuria/myoglobinuria.
- **Trauma, flank pain, clots** → urology + computed tomography.
- **Glomerular work-up**: C3, C4, antistreptolysin O/anti-DNase B, albumin → nephrology.
- **Macroscopic-only causes**: urethrorrhagia, tumour, nutcracker, renal vein thrombosis.

PROTEINURIA

- **UPCR normal**: <0.7 newborn, <0.2 mg/mg child; <4 mg/m²/h; adult <150 mg/day.
- **Transient** = >75% of isolated proteinuria (fever, exercise).
- **Orthostatic**: first-morning UPCR <0.2, raised upright; adolescents; benign.
- **Fixed** → full work-up; systemic → C3, C4, antinuclear antibody; normal screen → biopsy.

GLOMERULAR DISEASE

- **CNS (congenital nephrotic syndrome)**: <3 months; no immunosuppression.
- **Secondary nephrotic** ≈10%; ~30 genes in primary.
- **Steroid-sensitive (SSNS)**: remission ≤8 weeks; ~50% relapse repeatedly.
- **Frequently relapsing**: ≥2 relapses/6 months or ≥4/12 months.

- **Steroid-dependent**: 2 consecutive relapses on steroids.
- **SRNS**: 15–20%; no remission 6–8 weeks → biopsy.
- **Steroid-sparing**: cyclophosphamide, mycophenolate mofetil (MMF), rituximab, calcineurin inhibitors (CNIs).
- **Minimal change (MCNS)**: 90% <10 years, 70–80% overall; median 2.5 years; ~90% steroid remission.
- **FSGS**: <25% steroid-responsive; recurs post-transplant, graft loss.
- **Membranous**: rare, mostly secondary (SLE); primary → anti-PLA2R.
- **Post-streptococcal GN (PSGN)**: weeks after infection; rare <3 years; C3 <50%.
- **PSGN recovery**: oedema 5–10 d; BP 2–3 weeks; C3 normal 8–12 weeks.
- **PSGN biopsy** if persistently low C3 or atypical course.
- **MPGN/C3GN**: rare <5 years; biopsy required.
- **IgA nephropathy (IgAN)**: commonest primary GN; haematuria with upper respiratory infection (URTI); biopsy only definitive.
- **IgAN ESKD** 20–30%; progression: proteinuria, hypertension, ↑ creatinine.
- **Henoch-Schönlein purpura**: clinical diagnosis; relapses 4 months; abnormalities 2 years in 10–20%.
- **Alport**: X-linked COL4A5 mostly; deafness; ~50% males ESKD by 25.
- **Antineutrophil cytoplasmic antibody (ANCA) GN**: untreated 90% dead in 2 years; ~¼ ESKD.
- **Granulomatosis with polyangiitis vs microscopic polyangiitis**: upper-airway granulomas only in former.
- **Anti-glomerular basement membrane**: α3(IV) NC1; after shockwave lithotripsy or Alport transplant.
- **HUS** = common cause of severe AKI in healthy young child.
- **STEC+ HUS**: O157:H7 >70%; ~half dialysis; 5–25% later CKD.
- **aHUS** ~10%: complement defect; ADAMTS13 to exclude thrombotic thrombocytopenic purpura; relapses post-transplant.

TUBULOPATHIES

- **RTA**: normal anion gap acidosis, urine pH >5.5, positive urine anion gap.
- **Proximal RTA**: HCO₃⁻ 14–17 mEq/L; urine pH can fall <5.5.
- **Fanconi**: glucosuria, aminoaciduria, phosphaturia; cystinosis common cause.
- **Fanconi drugs**: cisplatin, ifosfamide, valproate.
- **Distal RTA**: hypokalaemia, hypercalciuria, nephrocalcinosis, rickets.

- **Type 4:** hyperkalaemia; CNIs, anti-inflammatories, obstruction.

CYSTIC RENAL DISEASE

- **Multicystic dysplastic kidney (MCDK):** non-communicating cysts; bilateral → lethal.
- **MCDK anomalies** up to 25% (vesicoureteral reflux, pelviureteric obstruction, dysplasia) → VCUG.
- **MCDK follow-up:** 3-monthly year 1 → 6-monthly to involution/≥5 years.
- **Autosomal recessive polycystic kidney disease (ARPKD):** PKHD1; 1:40,000; hepatic fibrosis; >75% ESKD by 5 years.
- **ARPKD neonatal mortality** 25–50% with oligohydramnios.
- **Autosomal dominant (ADPKD):** PKD1/PKD2; 1:1000; adult presentation usually.
- **MCDK long-term:** hypertension/CKD risk → annual review into adulthood.
- **Nephronophthisis:** polyuria, anaemia → ESKD early adolescence.
- **Senior-Løken** — retinitis pigmentosa; **Joubert** — vermian aplasia.
- **Bardet-Biedl** — obesity, polydactyly; **Jeune/Ellis-van Creveld** — skeletal.

HYPERTENSION

- **Hypertension:** repeated blood pressure (BP) >95th percentile; prevalence 3–4%.
- **Normal (American Academy of Pediatrics):** <90th percentile age 1–13; <120/80 age ≥13.
- **White-coat:** up to ¼.
- **Emergency** = end-organ injury; **urgency** = none.
- **<1 year:** coarctation or renovascular.
- **School age:** renal parenchymal (reflux nephropathy).
- **Obese >6 years, family history; adolescents** → essential.
- **No screening ultrasound >6 years** unless abnormal urinalysis/function.
- **Echo at drug start**, not at diagnosis.
- **First-line:** lifestyle (stage 1/primary, no end-organ damage).
- **Drugs if:** stage 2, end-organ damage, symptoms, secondary, lifestyle failure.
- **Target:** <90th percentile; <130/80 adolescents; CKD/diabetes <50th.

ACUTE KIDNEY INJURY

- **KDIGO (Kidney Disease: Improving Global Outcomes) stage I:** 1.5–1.9× or ≥0.3; UO <0.5 for 6–12 h.
- **Stage II:** 2.0–2.9×; UO <0.5 ≥12 h.
- **Stage III:** ≥3×, Scr ≥4.0, eCrCl <35; UO <0.5 ≥24 h or <0.3 ≥12 h.
- **Incidence:** ~4/1000 admissions; up to ¼ of intensive care admissions.
- **Mortality** 15% vs 0.6% without AKI.
- **Creatinine lags** 24–72 h; rises only after ~50% GFR loss.
- **Fractional excretion of sodium** <1% prerenal; >3% intrinsic; on diuretics urea <35%.
- **Prerenal:** vigorous isotonic fluid, no K⁺ until urine flows.
- **Early AKI:** dose drugs as GFR <15.
- **Renal replacement therapy if:** refractory overload, K⁺/metabolic, uraemia, nutrition need.
- **Gold standard modality:** CRRT (continuous renal replacement therapy).
- **AKI → CKD** in ~⅓ on long follow-up.

CHRONIC KIDNEY DISEASE

- **Chronic kidney disease (CKD):** abnormality >3 months; staging excludes <2 years.
- **Stages (GFR):** I ≥90, II 60–89, III 30–59, IV 15–29, V <15.
- **Young child causes:** CAKUT (posterior urethral valves, prune belly), cystic; adolescents: glomerular.
- **Height** ≈1.5 SD below; growth hormone (GH) resistance, not deficiency.
- **Malnutrition** up to 20%; **dyslipidaemia** 39–65%.
- **Anaemia:** erythropoietin deficiency → erythropoiesis-stimulating agent + iron.
- **Cardiovascular** ≈ half of deaths on dialysis.
- **Developmental problems** up to 25% of infants.
- **CKD mineral bone disease:** calcitriol, vitamin D₃, phosphate binders.
- **Protein restriction:** not recommended.
- **Only proven brake:** BP control → 50th percentile (KDIGO).

ESKD: DIALYSIS AND TRANSPLANTATION

- **Dialysis start:** GFR <15 or complications; no absolute creatinine.
- **Peritoneal dialysis (PD) first** <9 years and <20 kg.
- **PD contraindications:** omphalocele, gastroschisis, diaphragmatic hernia, bladder exstrophy.
- **Vesicostomy** does not preclude PD.
- **Haemodialysis** predominant >10 years; 3–4×/week, 4 h.
- **Goal = transplant;** pre-emptive if living related donor.
- **Absolute contraindications:** active malignancy, uncontrolled infection.
- **Native nephrectomy:** recurrent pyelonephritis, refractory hypertension, nephrotic syndrome.
- **Graft 50% survival:** living 20 years; deceased 12.5 years.
- **Delayed graft function:** 5% living, 15% deceased.
- **Year-1 rejection:** <10% living, ~15% deceased.
- **Maintenance:** tacrolimus/ciclosporin + MMF; steroid-free in low risk.
- **Primary graft dysfunction** <3%; induction thymoglobulin or basiliximab.
- **Malignancy** ≈7-fold (renal cell, thyroid, skin).

CLASSIC TRAPS

- **Trap:** day-1 creatinine = neonatal function — Actually: maternal until 48 h.
- **Trap:** image microscopic haematuria with VCUG/cystoscopy — Actually: not indicated.
- **Trap:** biopsy new childhood nephrotic syndrome — Actually: steroids first.
- **Trap:** immunosuppress congenital nephrotic syndrome — Actually: unresponsive; not recommended.
- **Trap:** IgAN haematuria weeks after URTI — Actually: concurrent; weeks = PIGN.
- **Trap:** platelets for STEC+ HUS — Actually: avoid unless bleeding/procedure.
- **Trap:** proximal RTA cannot acidify urine — Actually: pH <5.5 below threshold.
- **Trap:** CKD child is GH-deficient — Actually: GH-resistant.
- **Trap:** restrict protein in paediatric CKD — Actually: no benefit; BP control.
- **Trap:** intellectual disability excludes transplant — Actually: not automatically.